

Custom 3D Printed Wrist Brace Featuring a Voronoi Pattern

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Abstract

The design of this custom-fitted 3D printed wrist brace is meant to create a brace which is breathable, cost-efficient, customizable, and comfortable while still functioning as a brace for immobilization. Customized braces are not widely available on the market, specifically at an affordable price using freely available resources. This project serves as a method to create custom, affordable braces which are comfortable and functional for the user, supporting good health and well-being. The 3D printed wrist brace, featuring a Voronoi pattern, merges applied mathematics with creative design. It showcases the bridge between mathematical concepts and modern fabrication techniques resulting in a tangible piece of art.

CCS Concepts

• **Social and professional topics** → User characteristics; People with disabilities; • **Applied computing** → Physical sciences and engineering; Engineering; • **Computing methodologies** → Computer graphics; Shape modeling; Mesh geometry models; • **Human-centered computing** → Accessibility; Accessibility design and evaluation methods.

Keywords

Voronoi pattern, scanning, thermoplastic polyurethane (TPU), reusable, 3D printing

ACM Reference Format:

Ayushi Bhattacharjee, Sabrina Zhou, Kayla Lee, Jennifer R Weiser, and Mili Shah. 2025. Custom 3D Printed Wrist Brace Featuring a Voronoi Pattern. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720303>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3720303>

1 Introduction

In 2017, there were 179 recorded cases of hand and wrist fractures per 100,000 people worldwide, with the highest counts in Central Europe (667 per 100,000) and Australasia (653 per 100,000) [5]. These injuries affect all ages, genders, and nationalities. Although wrist injuries are not fatal, they result in chronic pain and reduce productivity and quality of life without proper management [10]. It costs on average 3.5k USD in both direct and indirect costs to treat these injuries per person. Currently, wrist braces are made using materials that are either too bulky or lack the rigidity needed to immobilize the injured area. Thus, there is a need to design and use a material for a lightweight, rigid brace that still gives the wearer the ability to do everyday tasks while offering a low-cost solution.

This project contributes to **good health and well-being**, a sustainable development goal of the United Nations. The purpose of the 3D printed wrist brace is to immobilize the wrist without further damaging the user's skin. This serves to improve well-being by allowing users to complete daily tasks (driving a car, taking a shower, playing sports) without causing further damage to the user's body. By creating an immobilizing brace, the user can participate in their normal activities by providing support for their injured body part. Furthermore, the proof-of-concept custom design process could be extended for adaptation in the future to other areas of the body (ankle, neck, etc.).

The 3D printed wrist brace project supports **responsible consumption and production** through a geometric pattern, the Voronoi pattern. This pattern creates the geometry of the brace, allowing for a more breathable and sustainable design by using less material without compromising rigidity. Furthermore, the material of the brace is thermoplastic polyurethane (TPU) which is both reusable and washable. By combining a reusable material and minimizing the surface area of the brace itself, responsible consumption and production goals are met. Additionally, all techniques used to create the project are freely available, allowing others to freely access, modify, and replicate the process. This fosters collaboration and innovation.

The scanning technique to create the .stl file for 3D printing the brace uses common technology accessible through smartphones, specifically the iPhone mobile digital device. The iPhone mobile digital device was chosen because it is a common device, with 1.5

billion Apple-branded iPhone mobile digital device products have been sold worldwide in 2018 [7]. Thus, the scanning technique to create the brace does not require the patient to own expensive scanning-specific equipment or travel to obtain a custom scan. A smartphone camera, a free app, and digital files can all be used at home and a brace can be 3D printed remotely, regardless of where the patient lives.

The motivation for this specific wrist brace is an individual who lost mobility of their left wrist after being involved in a vehicular accident. Consequently, they were seeking a breathable, customized wrist-brace that would allow them to complete daily tasks such as emptying the dishwasher, cleaning the house, and taking care of their newborn children. The challenge was the individual was not located in the same geographic location as the design team. Over the course of two years, our team worked together to perfect the process and maximize efficiency, while not being able to physically interact with the individual in need.

There are other wrist-immobilization projects currently being researched. For instance, there is a patient-specific wrist splint for patients with rheumatoid arthritis [12]. The process, while similar to ours, differs slightly in each step. Our scanning process uses a holistic 3D scan, while the method for rheumatoid arthritis brace stitches together four images. Additionally, we use TPU while the rheumatoid arthritis brace uses a combination of PLA and resin [12]. Thus, our method is faster and less expensive. There is also ActivArmor™, which offers 3D printed waterproof casts and splint that are “available to athletes, physicians, and clinics worldwide” [1].

2 Methodology

The original inspiration for the brace is ActivArmor™, a custom 3D printed support device designed for personalized comfort and support [1]. The goal of the project was to develop an in-house technique to create a similar brace using only freely accessible software, making the process accessible and cost-effective. This approach aimed to provide a customizable, easily reproducible alternative for medical or personal use featuring a geometric Voronoi pattern to reduce material costs and increase breathability.

The production of the 3D printed wrist brace uses 3 main forms of technology. First, the app Scandy™ Pro, available for free on iOS and Android everyday cell phones, is used to create a custom scan of the user's wrist (or any body part) and exported as an .stl file. Then, the .stl file is imported into the Autodesk® software Meshmixer to create a mesh ready for 3D printing [8]. Scanning defects are erased and smoothed, the mesh densities are altered to create a solid mesh around stabilizing parts (the outline, the opening slit for putting on the brace, and the thumb hole). Then, the rest of the brace is converted into a Voronoi pattern, to maximize breathability while minimizing material used and therefore decreasing the cost of material needed. Once the mesh is completed, the .stl file is scaled to the size of the user's hand and 3D printed on a Bambu Lab X1C 3D Printer with TPU filament [4]. The filament was Bambu Lab brand TPU 95A HF [3].

The 3D printing settings chosen were a speed of 20 mm/s, filament temperature of 225°C, and 20% infill to decrease the chance of

defects, ridges, and/or bumps that would create a comfortable and safe brace to use against the skin for the user.

2.1 Scanning: Using Freely Available Software to Create Scans

The software used for creating the customized scan of the client's wrist is Scandy™ Pro, a free app compatible with iOS Apple devices, which exports files in .stl format to Autodesk® Meshmixer [8]. During the scanning process, the client (on the right) holds their wrist straight and upright while a second person (on the left) rotates the camera 360° around the arm, as seen in Fig. 1. The file is then cropped and scanning noise is discarded before exporting as an .stl file. This technique was found to be the most accurate and efficient over the course of many trials. Although Scandy™ Pro was used for this project, there are other scanning apps available on smartphones which can be used to create an .stl file of the wrist.

2.2 Autodesk® Meshmixer: Creating the Voronoi Pattern

The Voronoi pattern occurs naturally throughout nature due to its efficiency and optimization of space to minimize energy and resources used [9]. The principal behind the pattern is that every point within a given region is closer to the “seed” inside that region than it is to any other point outside that region. Additionally, each point along a region's edge is equidistant from the two nearest seeds.

Consider a set S of three or more points in the Euclidean plane, not all collinear, and that no four points are cocircular. Suppose all these points radially expand until they all meet at straight lines, as visualized in Fig. 2. From this set, take an arbitrary pair of points P and Q . The bisector of the line joining P and Q (the red line in Fig. 2) is the locus of points equidistant from both P and Q . This is true for any points P and Q , optimizing space within the boundary [9].

Fig. 3 (1), shows the scan imported into the free Autodesk® software Meshmixer. Once the .stl file is imported, there are slight imperfections that must be smoothed before the mesh can be sent to be 3D printed. Fig. 3 (2) shows an evolution of this process for a wrist brace. First, the density of the mesh is reduced by 50-200%. The first four fingers are removed by implementing a plane cut and smoothing the boundary to prevent any sharp edges. This can be seen as the green portion of the mesh. Then, the thumb is selected (orange) and removed, followed by smoothing the boundary edges.

After the thumb hole is created, the density of the mesh is altered to create the Voronoi pattern. This is seen in Fig. 3 (3), where the denser regions represent the solid, structural part of the brace and the wider triangles represent the blueprint for the Voronoi pattern. Once the densities are altered accordingly, the brace is offset by 0.145 inches to create the mesh from which the 3D model will be created. The 0.145 inch offset allows the brace to comfortably fit the user's hand and wrist while still providing structural support. If the mesh were not offset, the brace would constrict the wrist. The mesh is then selected, and the pattern “Dual Edges” in Meshmixer is applied to the model. The brace is given a thickness of 0.28 inches, optimized to provide support without being bulky or too heavy. The resulting 3D model is seen in Fig. 3 (4).



Figure 1: Scanning the arm using Scandy™ Pro.

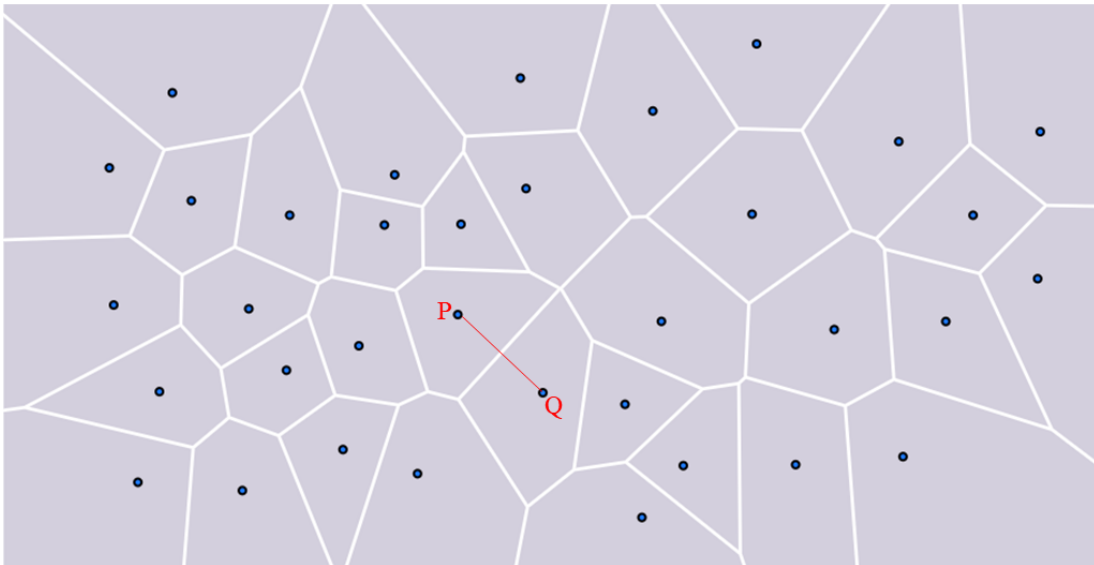


Figure 2: Voronoi pattern defined by 35 points [11].

2.3 3D Printing using Thermoplastic Polyurethane (TPU)

Thermoplastic polyurethanes (TPUs) are created via the polymerization of a polyol with a diisocyanate [6]. As a result, the material properties include modifiable flexibility, mechanical strength, and abrasion resistance. These characteristics align with the goals of this project, as the brace needs to be sturdy for immobilization

but flexible for comfort and breathability. Furthermore, abrasion resistance allows for the life of the brace to be extended, allowing the user to wear the brace for multiple years without compromising structural stability [6]. Fig. 4 shows the typical repeat structures for TPU. Softer segments contain longer chain, aliphatic diols for more flexibility alternating with harder segment repeat units that

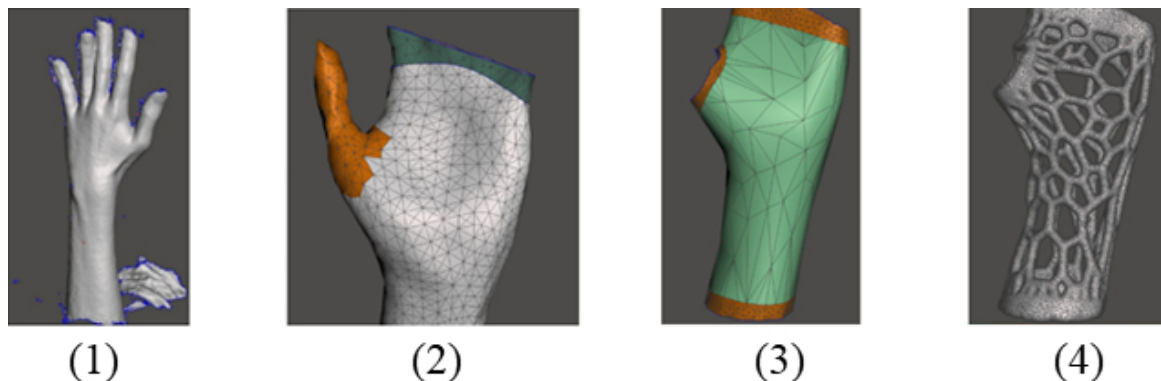


Figure 3: (1) The scan imported into Meshmixer as an .stl file. (2) Step 1 of the Meshmixer process. (3) Altered mesh to create Voronoi pattern. (4) 3D version of brace ready for 3D printing

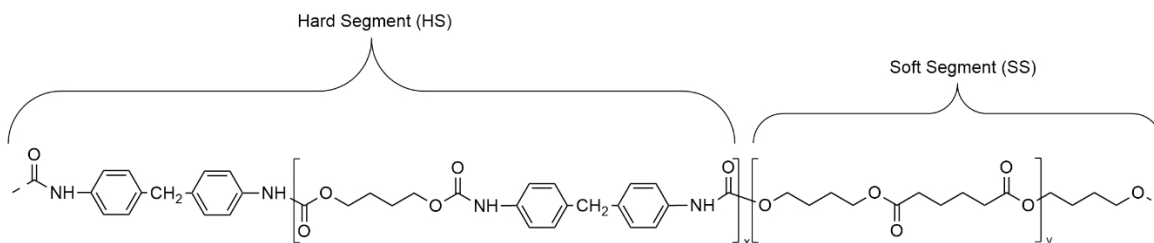


Figure 4: Typical repeat unit structures for thermoplastic polyurethane (TPU).

are shorter chain aliphatic diols attached to diisocyanates to reduce flexibility and improve mechanical strength.

Furthermore, TPU was chosen due to being inherently waterproof and oil-resistant [2]. This quality is important for daily activities, where the user will encounter these liquids. This allows the user to participate in activities involving water, such as swimming, cooking, or showering, and not be compromised by daily exposure to the oils produced in skin. If the brace can resist these fluids, the life of the brace extends, and the user will not have to frequently buy new braces.

A benefit to selecting TPU is the ready availability of the material and the increasing demand for TPU in the market. The TPU market is estimated to grow 6-7% by 2025 [2]. Lastly, TPU is reusable, promoting *responsible consumption and production*, a goal of the design brief, with a price of only 42 USD per kilogram [3].

2.4 Completed 3D Printed Wrist Brace

The completed brace can be seen in Fig. 5. (Left). Cylindrical notches are present on both sides of the clamshell opening and connected across by O-rings to create a closing mechanism. This can be seen in Fig. 5 (Right). This closing mechanism provides a simple, efficient method for the user wearing the brace to put it on without help. This takes into consideration that user may only have one hand to put on and close the brace around their wrist, as is the case with the specific individual for whom our brace was developed.



Figure 5: (Left) 3D Printed Custom Brace with Thermoplastic Polyurethane. (Right) Clamshell opening of brace with O-rings used in early iterations.

The brace takes, on average, one day to design and 8-12 hours to 3D print, provided there are no printer malfunctions. Typical 3D printer malfunctions were encountered as the fabrication process was optimized. These malfunctions included filament getting stuck in the nozzle, ridges and bumps in the brace itself, a stringy finish due to TPU material properties, or the print failing mid-way. However, 3D printing techniques were optimized over the course



Figure 6: Chronological iterations of the wrist brace.

of two years to create a smooth brace, removing the need for post-processing. Optimizing the printing techniques included a slower nozzle speed of 20 mm/s, filament temperature of 225°C, and 20% infill.

The Bambu Lab X1 Carbon Printer, available for only 1000 USD, was chosen as the machine used to optimize the 3D printing process for the brace [4]. The Bambu Lab X1 Carbon Printer was found to create the smoothest braces minimizing the need for post-processing compared to other inexpensive, commercially available 3D printers. This makes the 3D printing of braces more easily repeatable for future iterations and enables an easier scale-up of the process.

3 RESULTS

The process and methodology of creating the 3D printed wrist brace has been refined throughout the design iterations, incorporating user feedback, post-processing and material testing, and optimization of the Voronoi pattern for improved comfort and functionality. Each iteration has provided valuable insights and enhancements in fit, materials used, support, and breathability.

3.1 Iterations of the 3D Printed Wrist Brace

Fig. 6 shows the evolution of the 3D printed wrist brace. The first iteration (a) is a standard wrist brace, providing support but lacking flexibility. As a result, it was not comfortable for the user and left indentations on their skin. The second iteration (b) showcases the opposite problem – a brace that is flexible, breathable, and fitted but does not provide wrist support. The last iteration (c) shows a brace that solves both issues, creating a brace that is flexible, breathable, and stable.

Various individuals at our institution have tested the brace and reported that it keeps the wrist rigid, and the custom-fit allows a comfortable, lightweight fit, especially on the pisiform bone of the

wrist. One individual remarked the brace was supportive but not constraining and molded to their wrist snugly. Additionally, the original patient for this brace reported they were able to more easily complete daily tasks with the brace, and they preferred it over a market-available brace. They reported the brace was breathable, easy to use, washable, and immobilizing.

The individual who served as the catalyst for the 3D printed wrist brace project was sent iterations of the brace, and was involved in the decision-making process for sizing, shape, thickness, etc. Due to their lack of feeling in the afflicted wrist, the individual was initially sent a custom-made brace for the opposite wrist to test for any roughness/bumps, validate the length of the brace, and give opinions on future goals and overall reviews. After this meeting, their requests were taken into account, and the design was altered accordingly.

The brace closure mechanism was changed from O-rings to Velcro to allow for size adjustability; for instance, if the user wanted to wear a sleeve under the brace, the Velcro would allow the brace to be worn without compromising comfort. Both closing mechanisms can be used on the brace depending on the user's preference. The length of the brace was shortened 0.25-0.75 inches to decrease the weight and bulk of the brace without compromising wrist immobility. In a subsequent meeting, after a custom-made brace was sent to them, the user reported they felt no bumps or scratches. Furthermore, with their wrist immobilized, they stated they were able to better complete daily tasks and take care of their newborn baby in a better manner than previously without the brace. They also mentioned the tolerance was perfect, as the brace did not dig into their skin nor was too loose to serve its purpose. Additionally, the user liked that the brace is waterproof, meaning they can wear it in the shower, when washing their hands, or for any other task where water was involved. The user wearing the brace with a sleeve



Figure 7: User wearing 3D printed wrist brace with a sleeve underneath.

underneath is shown in Fig. 7, their preference as to prevent any unintentional scratching as they have no feeling in their wrist.

3.2 Final Iteration of 3D Printed Wrist Brace

The ease of scanning and producing 3D printed braces makes fabricating these braces for other parts of the body simple. Work on braces for the leg and neck has begun, seen in Fig. 8. By relying solely on free software, we have ensured that these methods are both accessible and reproducible for other researchers and developers. In fact, a research group at our institution is currently adapting Scandy™ Pro to design customized exoskeletons for first responders, leveraging the precision and flexibility of this approach for a new application. This adaptability highlights the potential of freely available solutions to support diverse projects, from medical devices to wearable support systems, encouraging innovation without the high cost typically associated with proprietary software.

4 CONCLUSION

The 3D printed brace showcases knowledge of the materials and techniques. A blend of open access software, commonly available materials, and 3D printing was used to develop this new way of producing immobilization braces. On the software-side, the free app, Scandy™ Pro, made it easy for an individual to create 3D scans of their own body for post processing, even from different geographical locations. With access to a smartphone, any individual can create a scan of their wrist, contributing to patient accessibility and thus *good health and well-being*.

The conversion of those scans into a Voronoi pattern helps create a lightweight, but strong brace. Furthermore, the Voronoi pattern captures attention with its unique geometry, harnessing aspects of applied mathematics towards an artistic creation. Its aesthetic makes it a strong contender across all evaluation criteria, highlighting a blend of mathematics and wearable art. On the hardware-side,



Figure 8: Voronoi pattern 3D printed braces for the knee, neck, and wrist (left to right).

different 3D printing settings were tested and optimized to achieve the smoothest brace, which would not scratch or irritate the wearer to allow for long-term use. TPU filament was the material chosen as it lends durability and flexibility to the brace while allowing intricate detailing, showcasing the Voronoi pattern to minimize materials and cost.

The brace design has been showcased at an end-of-semester show at the university. Many different sizes and iterations were displayed, allowing guests of various hand sizes to wear the braces. They all had similar reviews, agreeing that there are limited braces on the market, currently, which are both comfortable and functional. Many noted that they would be interested in wearing this brace for their own injuries. They mentioned that not only did the Voronoi pattern look aesthetically pleasing, but the flexibility it gave the brace while still being durable was impressive and innovative. Thus, overall, the design was validated for its design goals.

The brace was also showcased at the Bridges Richmond 2024 Conference, an interdisciplinary conference bridging together mathematics and art [13]. A model wore 3D printed neck, waist, wrist, and leg braces and walked the fashion runway, showcasing the aesthetic, geometric Voronoi pattern and functionality of the brace. The unique geometry portrays mathematical principles while the pattern makes a lasting impression, showcasing the relationship between mathematics and aesthetics. Additionally, the pattern allows minimal TPU to be used, and TPU itself is reusable, contributing to *responsible consumption and production*.

Our custom 3D printed wrist brace allows a solution for creating braces regardless of the patient's location, utilizing accessible and relatively affordable software and hardware. With access to a smartphone, any individual can use the free app Scandy™ Pro to create a scan of their wrist, which can be sent to our team remotely via an .stl file. This file can then be edited and remeshed in the free software Autodesk® Meshmixer [8]. Once the Voronoi pattern is applied, the brace is ready to be printed with TPU, with each brace

costing around 5-15 USD. Once the print is complete, the custom wrist brace features an aesthetic, geometric pattern with minimal material costs.

Currently, our team is working on refining the methodology and creating an app freely available to the public that is fully open-sourced. By doing so, the patient does not need to be physically near a 3D printer to receive a custom brace, should the individual not have access, increasing inclusivity and accessibility. Future stakeholders can use this to their advantage as it reduces the cost of software and eliminates proprietary barriers enabling easier integration into healthcare systems worldwide.

Acknowledgments

Many thanks to our mentors Dr. Mili Shah, Professor and Chair of Mathematics at The Cooper Union for the Advancement of Science and Art., and Dr. Jennifer Weiser, Associate Professor of Chemical Engineering at The Cooper Union for the Advancement of Science and Art. Thank you to a private bioengineering grant from a foundation that wishes to remain anonymous that helped fund the project.

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